THE LIFTING CATEGORY IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1018-1022 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1024-1096, the complete author proof (71 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_stacks-stacks-more-morphisms.json (pb=3497, stmt 1018-1022, proof 1024-1096).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: descent of morphisms left to the reader and transitivity of the equivalence 'omitted' - mechanism-named routine bookkeeping (artin precedent).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-gerbe-of-lifts-fibred: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 964–967. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-morphisms-lifts-etale: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 950–952. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BPG: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1096–1096. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-gerbe-of-lifts: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 875–903. Cited by the retained

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statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02WW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1096–1096. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02ZI: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1096–1096. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 026C: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1096–1096. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The category $p : \mathcal{C} \to W_{spaces,étale}$ constructed in Remark remark-gerbe-of-lifts is a stack in groupoids.*

Proof. By Lemma lemma-gerbe-of-lifts-fibred we see the first condition of Stacks, Definition 02ZI holds. As is customary we check descent of objects and we leave it to the reader to check descent of morphisms. Thus suppose we have $a : U \to W$ in $W_{spaces,étale}$, a covering $\{U_k \to U\}_{k\in K}$ in $W_{spaces,étale}$, objects $\xi_k = (U_k, U'_k, a_k, i_k, x'_k, \alpha_k)$ of $\mathcal{C}$ over U_k, and morphisms

$$\varphi_{kk'} = (f_{kk'}, f'_{kk'}, \gamma_{kk'}) : \xi_k|_{U_k\times_U U_{k'}} \to \xi_{k'}|_{U_k\times_U U_{k'}}$$

between restrictions satisfying the cocycle condition. In order to prove effectivity we may first refine the covering. Hence we may assume each U_k is a scheme (even an affine scheme if you like). Let us write

$$\xi_k|_{U_k\times_U U_{k'}} = (U_k \times_U U_{k'}, U'_{kk'}, a_{kk'}, x'_{kk'}, \alpha_{kk'})$$

Then we get an étale (by Lemma lemma-morphisms-lifts-etale) morphism $s_{kk'} : U'_{kk'} \to U'_k$ as the second component of the morphism $\xi_k|_{U_k\times_U U_{k'}} \to \xi_k$ of $\mathcal{C}$. Similarly we obtain an étale morphism $t_{kk'} : U'_{kk'} \to U'_{k'}$ by looking at the second component of the composition

$$\xi_k|_{U_k\times_U U_{k'}} \xrightarrow{\varphi_{kk'}} \xi_{k'}|_{U_k\times_U U_{k'}} \to \xi_{k'}$$

We claim that

$$j : \coprod\nolimits_{(k,k')\in K\times K} U'_{kk'} \xrightarrow{(\coprod s_{kk'}, \coprod t_{kk'})} (\coprod\nolimits_{k\in K} U'_k) \times (\coprod\nolimits_{k\in K} U'_k)$$

is an étale equivalence relation. First, we have already seen that the components s, t of the displayed morphism are étale. The base change of the morphism j by $(\coprod U_k) \times (\coprod U_k) \to (\coprod U'_k) \times (\coprod U'_k)$ is a monomorphism because it is the map

$$\coprod\nolimits_{(k,k')\in K\times K} U_k \times_U U_{k'} \longrightarrow (\coprod\nolimits_{k\in K} U_k) \times (\coprod\nolimits_{k\in K} U_k)$$

Hence j is a monomorphism by More on Morphisms, Lemma 0BPG. Finally, symmetry of the relation j comes from the fact that $\varphi_{kk'}^{-1}$ is the "flip" of $\varphi_{k'k}$ (see Stacks, Remarks 026C) and transitivity comes from the cocycle condition (details omitted). Thus the quotient of $\coprod U'_k$ by j is an algebraic space U' (Spaces, Theorem 02WW). Above we have already shown that there is a thickening $i : U \to U'$ as we saw that the restriction of j on $\coprod U_k$ gives $(\coprod U_k) \times_U (\coprod U_k)$. Finally, if we temporarily view the 1-morphisms $x'_k : U'_k \to \mathcal{X}'$ as objects of the stack $\mathcal{X}'$ over U'_k then we see that these come endowed with a descent datum with respect to

the étale covering $\{U'_k \to U'\}$ given by the third component $\gamma_{kk'}$ of the morphisms $\varphi_{kk'}$ in $\mathcal{C}$. Since $\mathcal{X}'$ is a stack this descent datum is effective and translating back we obtain a 1-morphism $x' : U' \to \mathcal{X}'$ such that the compositions $U'_k \to U' \to \mathcal{X}'$ come equipped with isomorphisms to x'_k compatible with $\gamma_{kk'}$. This means that the morphisms $\alpha_k : x \circ a_k \to x'_k \circ i_k$ glue to a morphism $\alpha : x \circ a \to x' \circ i$. Then $\xi = (U, U', a, i, x', \alpha)$ is the desired object over U. □